Aiyman Akbar
Annotated Bibliography

1) Ernst, Ekkehardt, Rossana Merola, and Daniel Samaan. "Economics of artificial intelligence: Implications for the future of work." IZA Journal of Labor Policy 9.1 (2019). https://sciendo.com/article/10.2478/izajolp-2019-0004

This paper talks about the implications of AI in social settings like task automation, jobs, wages, the emergence of new tasks and new skills, and the specifics of AI in economic context. This paper discusses the potential positives and negatives of different topics, making it useful towards my research and the questions I want to answer. It offers a balanced view of AI in an economic context, answering questions like the types of jobs at risk of AI automation and the new skills that will be required for these AI-influenced work environments, and even discusses the role of policy in the management of AI's impact on the economy, which will provide me with interesting insights to put into my research paper.

This paper uses both case studies and qualitative analysis when analyzing the economics of AI. Some of the qualitative data includes labor market data and industry reports in the forms of graphs, charts and citations, and case studies such as:

- The paper discusses how AI has replaced certain manual tasks such as those in manufacturing plants while also creating new technical jobs such as maintenance and programming. This case study examines the balance between job displacement and job creation in the manufacturing industry due to AI.
- This case study explores the use of AI in standard medical procedures and diagnosing patients, as well as displacement of traditional medical roles like radiologists while also promoting new opportunities in AI development and maintenance, much like the previous case study. It shows how AI can enhance productivity and accuracy in medical diagnostics, specifically with new AI tools like algorithms capable of medical imaging..
- The paper discusses a cause study about the implementation of AI systems that are more efficient and accurate in financial markets and during trading and how they pose challenges to traditional trading jobs. This case explores the between the benefits of AI in increasing productivity and the risks of people losing their jobs due to this increased productivity.
- This case study examines the shift from traditional customer service roles to AI roles like chatbots and personalized AI marketing algorithms, looking into how AI can improve customer experience and the effect that will have on jobs in the retail market.

This paper concludes that AI is a fantastic tool for revolutionizing productivity, efficiency and

time management, especially in routine based tasks like manufacturing. However, the drawback discussed in the paper is that this will affect the jobs of the people who are in those roles currently. The paper concludes with the importance of proper policy implementation to battle this job displacement, such as investing in teaching the newer generations the skills that will be required in an AI filled world. The article also talks about how proper management of AI is crucial to avoid making issues like wage inequality worse.

2) Cao, Longbing. "AI in finance: A review." Available at SSRN 3647625 (2020). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3647625

This paper provides a broad view of AI in finance and its development and potential in the future. It is a lengthy paper that discusses the role of AI in finance and both classic and modern AI techniques, outlining how data science, machine learning and AI enhance financial efficiency. This paper asks questions like How does AI transform financial modeling and economic simulations? What are the roles of AI in risk management and regulatory compliance? How can AI optimize investment strategies and market analysis? This kind of analysis of AI in the financial sector will be useful in my research when discussing AI in the financial sector of the global economy, specifically in regards to AI's strategic integration into financial systems, the economic benefits of AI integration and the potential challenges and risks.

This paper uses a mixed approach where quantitative and qualitative sources and pieces of information are used, as well as the usage and discussion of both traditional and more modern AI research in finance. It ranges from the discussion of the earliest financial computational models to more modern machine learning and deep learning finance models. Some case studies discussed include AI-driven risk management systems in banks, revolutionary automated trading algorithms in hedge funds, and the growing trend of customer service chatbots in retail banking. The paper also goes into a discussion about different AI methodologies, such as unsupervised vs supervised learning, reinforcement learning and natural language processing, discussing how each model and methodology could and can be applied in a financial context.

This paper finds that AI has and will continue to have a transformative impact on finance, in areas like enhancing efficiency, accuracy, and innovation. The paper also goes to talk about how there is a need for more research and innovation in regards to AI in finance (and AI in general) specifically in regards to proper regulation, ethics and security. The paper concludes with the idea that you need a balanced approach to maximize the benefits of AI and mitigate the most amount of risk as possible.

3) Buiten, Miriam, Alexandre De Streel, and Martin Peitz. "The law and economics of AI liability." Computer Law & Security Review 48 (2023): 105794. https://www.sciencedirect.com/science/article/pii/S0267364923000055

This paper breaks down the liability issues in regards to AI. It talks about the issues AI currently has on existing liability issues associated with the producers and the users. The paper also talks about how the existing rules should be changed to adapt to the new issues AI poses. It then analyzes the implications of different liability frameworks in regards to AI such as the Ai liability Directive and a proposed Liability Directive in the EU. This paper can be useful in my research in the introduction when I am explaining AI, as liability is a fundamental part of AI in the global economy and it should be discussed. Otherwise it can be useful in understanding the development of policies regarding AI, which can be useful in other parts of my research.

The methodology used in this paper is more legal document heavy and based on legal analysis with economic principles integrated to link to the liability issues regarding AI. It also includes a sprinkle of case studies here and there, as well as some historical review of liability laws of emerging technologies and how they have evolved. Some examples of the legal documents discussed in this paper in regards to liability are regulatory proposals and specified case law. Some of the case studies discussed in this paper are real-world examples of AI failures, such as autonomous vehicle accidents and AI-driven medical errors, which show the complex nature of liability for AI.

Some of the questions that this paper brings up and answers are: what are the challenges in applying existing liability rules to AI systems? How should liability be allocated between producers and users of AI? What changes are needed in liability laws to address the unique risks posed by AI?

This paper concludes that the current liability policies and frameworks are not enough to deal with the complex liability issues that come with AI. This is especially noticeable when trying to prove or justify things like fault and causality when it comes to AI. The paper suggest numerous policy recommendations for the EU in specific, such as strict liability and the discussion of the proposed EU AI liability directive, which the paper decides is a step in the right direction but is not refined or vast enough to effectively cover the liability issues of AI.

4) Jejeniwa, Temitayo Oluwaseun, Noluthando Zamanjomane Mhlongo, and Titilola Olaide Jejeniwa. "AI solutions for developmental economics: opportunities and challenges in financial inclusion and poverty alleviation." International Journal of Advanced Economics 6.4 (2024): 108-123. https://fepbl.com/index.php/ijae/article/view/1073

This paper talks about AI's potential in transforming developmental economics. Specifically the paper talks about how AI's ability in personalized data collection and risk assessment can amplify poverty alleviation and increase financial inclusion in an economy. The paper continues on about the abilities of AI to revolutionize development economic strategies but also talks about challenges that come with AI such as ethics. This paper will be extremely useful in the section where I discuss development economics and will allow me to analyze both the positives and negatives in terms of AI in development economics through case studies, best practice recommendations and examples.

This paper uses mixed methods in its research and analysis, including theoretical analysis of economic theories in regards to AI's role in development economics, case studies, and best practices and recommendations. Some of the case studies discussed are:

- The paper discusses the success of M-Pesa, a mobile banking solution in Kenya, which has significantly improved financial inclusion by providing easy access to financial services for the unbanked population.
- The paper talks about AI-driven finance initiatives that use machine learning algorithms to assess people's credit score and other factors in getting a loan and provide loans to small business owners India, improving their access to capital and supporting economic growth.
- The paper discusses the implementation of AI-based platforms that offer tailored financial products to farmers based on predictive models of crop yields and market prices. This helps in stabilizing incomes and promoting sustainable agricultural practices.

5) Korinek, Anton. "Generative AI for economic research: Use cases and implications for economists." Journal of Economic Literature 61.4 (2023): 1281-1317. https://www.aeaweb.org/articles?id=10.1257/jel.20231736 https://www.newyorkfed.org/medialibrary/media/research/conference/2023/FinTech/400p m_Korinek_Paper_LLMs_final.pdf

This is a lengthy paper that goes extremely in depth into generative AI and its potential in terms of economics research. It starts by talking specifically about LLM, a type of generative AI like chatGPT, and then talks about how these trained AI models can be applied to revolutionize economic research. The paper then goes extremely in depth about many different generative AI models that are popularly used currently and breaks down how they work and how they are and

can be used from an economic view. This paper can be useful in my research by providing countless examples of how AI can be integrated into economic research, including how it works and research methods. This AI can also be used to help make economic policy recommendations and create forecasts for financial institutions and countless other possibilities, which I can discuss in relation to the other sections of my research.

6) Furman, Jason, and Robert Seamans. "AI and the Economy." Innovation policy and the economy 19.1 (2019): 161-191. https://www.journals.uchicago.edu/doi/full/10.1086/699936

This source is a great analysis of AI and its effect on the economy through many different methods, but all revolving around AI's ability to increase productivity in an economy. This paper goes into how to adapt policy to best utilize the power of AI on labor markets and to reduce disruptions in labor markets. This paper's analysis of AI's potential to increase productivity in labor markets and analysis of regulation and policy strategies to balance the pros and cons of AI will be useful in my research when I talk about AI in the global economy both on a macro and micro scale. It can also be useful when talking about the international and developmental side as labor markets are present in these economies as well.

7) McKinsey & Company. "The Economic Potential of Generative AI: The next Productivity Frontier." McKinsey Digital, 14 June 2023. https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-economic-pote ntial-of-generative-ai-the-next-productivity-frontier#introduction

This source is McKinsey's official analysis of the potential of AI Across many sectors of the global economy. It goes through various different use cases, examples and sectors that AI can be applied to from an economic standpoint and also discusses the numerical value of how much AI could add to the global economy in the future. This source will be very useful in my research as it is very current and unlike some of my other sources is not an academic paper, it is an actual analysis by the leading digital consulting company in the world. It will give me many examples and use cases that I can incorporate in various parts of my research.

8) The White House. “FACT SHEET: President Biden Issues Executive Order on Safe, Secure, and Trustworthy Artificial Intelligence.” The White House, 30 Oct. 2023. https://www.whitehouse.gov/briefing-room/statements-releases/2023/10/30/fact-sheet-president-biden-issues-executive-order-on-safe-secure-and-trustworthy-artificial-intelligence/

This is the fact sheet for an executive order issued by president Biden in 2023 about new regulations and standards for AI in the US, in regards to things like privacy, safety and innovation. This source will be useful in my research when discussing ethics in regards to AI throughout the various sections of my research. More than that, however, this source will be interesting when I discuss the potential effects of nationalization of AI, and a question I have that I am eager to explore deeper. Based on the regulations and standards that president Biden puts in place for AI, I can look into the effects they had on the economy and hypothesize on what further shifts could occur in the US economy and the global economy due to the United States regulations on AI.